

PHASE

TO MAINTAIN THE ORIGINAL STRUCTURE OF A SIGNAL IN THE PASS-BAND OF A FILTER, (CLOSE TO) LINEAR PHASE IS REQUIRED $\Rightarrow$ PHASE IS STRAIGHT LINE.

↑
NOT REALLY INTERESTED
IN OTHER BANDS

A SYSTEM HAS LINEAR PHASE IF

$$\angle H(e^{j\omega}) = -c\omega \quad \forall \omega, \quad c \in \mathbb{R}$$

IN GENERAL A LINEAR PHASE SYSTEM HAS A FREQUENCY RESPONSE:

$$H(e^{j\omega}) = |H(e^{j\omega})| e^{-j\omega c}$$

GROUP DELAY

TO MEASURE THE LINEARITY OF THE PHASE OF A SYSTEM, WE CAN EVALUATE THE GROUP DELAY

$$\tau(\omega) = \text{grd}\{H(e^{j\omega})\} = -\frac{d}{d\omega} \arg\{H(e^{j\omega})\}$$

THE GROUP DELAY IS THE SAMPLE DELAY OF A SPECIFIC FREQUENCY. IF IT'S CONSTANT, ALL FREQUENCIES ARE DELAYED BY THE SAME NUMBER OF SAMPLES $\Rightarrow$ SAMPLE DELAY.

A LINEAR PHASE RESULTS IN A CONSTANT GROUP DELAY AND WE CAN MEASURE THE NON-LINEARITY BY LOOKING AT HOW MUCH THE GROUP DELAY DEVIATES FROM A CONSTANT VALUE.

CAUSAL FIR FILTERS

$$H(z) = \sum_{n=0}^M h[n] z^{-n} \quad \longleftrightarrow \quad H(e^{j\omega}) = \sum_{n=0}^M h[n] e^{-j\omega n}$$

FINITE LENGTH
STARTS AT 0

FIR FILTERS WITH PROPERLY CHOSEN SYMMETRIC OR ANTISYMMETRIC IMPULSES GUARANTEE LINEAR PHASE / CONSTANT GROUP DELAY.

THIS SIMPLIFIES THE DESIGN OF FIRs BY OPTIMIZATION TECHNIQUES

DEPENDING ON THE TYPE OF SYMMETRY AND IF M IS ODD OR EVEN, THERE ARE 4 TYPES OF LINEAR PHASE FIR FILTERS:

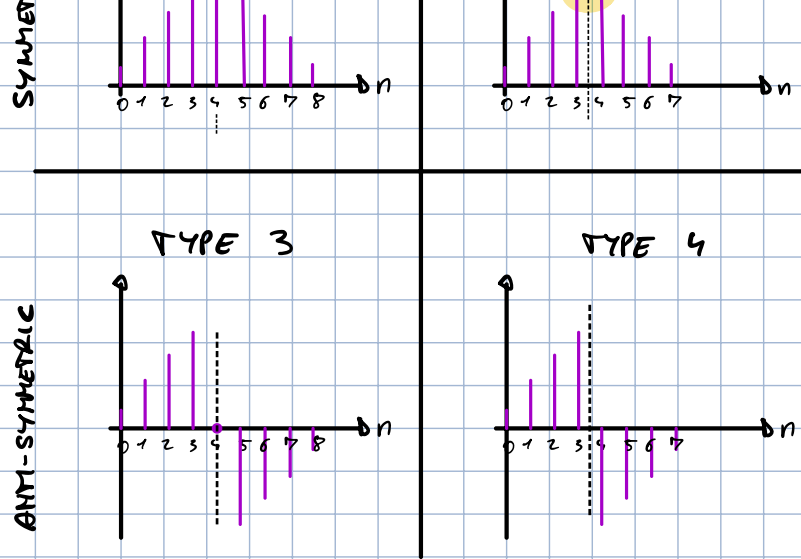
$$h[n] = \pm h[M-n], \quad M \in \mathbb{Z} \text{ ODD OR EVEN}$$

SYMMETRY

WHERE M IS THE ORDER OF THE FILTER

$$\hookrightarrow L = M+1$$





GENERALIZED DFT

IN THE CASE OF COSINE SYMMETRIC FIR FILTERS (TYPE 1, 2), WE CAN GENERALLY EXPRESS THE DFT AS

$$H(e^{j\omega}) = A(e^{j\omega}) e^{-j\omega \frac{M}{2}}$$

DIFFERENTLY IN THE CASE OF ANTI-SYMMETRIC FILTERS, WE INTRODUCE A CONSTANT TO THE PHASE TERM

$$H(e^{j\omega}) = j A(e^{j\omega}) e^{-j\omega \frac{M}{2}} = A(e^{j\omega}) e^{-j\omega \frac{M}{2} + j\frac{\pi}{2}}$$

↓
 $e^{j\frac{\pi}{2}}$

TYPE 1

$M=4, L=5$

ODD LENGTH + SYMMETRIC

$$h[n] = \{h[0], h[1], h[2], h[3], h[4]\}$$

↓ USE SYMMETRY

$$h[n] = \{h[0], h[1], h[2], h[1], h[0]\}$$

↓ DFT

$$H(e^{j\omega}) = \sum_{n=0}^M h[n] e^{-j\omega n} = h[0] + h[1] e^{-j\omega} + h[2] e^{-2j\omega} + h[1] e^{-3j\omega} + h[0] e^{-4j\omega}$$

$$= h[0] (1 + e^{-4j\omega}) + h[1] (e^{-j\omega} + e^{-3j\omega}) + h[2] e^{-2j\omega}$$

↓ WANT FORM REPRESENTABLE BY COSINES

$$= e^{-2j\omega} h[0] (e^{2j\omega} + e^{-2j\omega}) + e^{-2j\omega} h[1] (e^{j\omega} + e^{-j\omega}) + h[2] e^{-2j\omega}$$

$$= \underbrace{[h[0] (e^{2j\omega} + e^{-2j\omega}) + h[1] (e^{j\omega} + e^{-j\omega}) + h[2]]}_{:= A(e^{j\omega})} e^{-2j\omega}$$

REPRESENTS LINEAR PHASE OF THE FILTER

$$= [h[0] \cdot 2 \cos(2\omega) + h[1] \cdot 2 \cos(\omega) + h[2]] e^{-2j\omega}$$

$$= A(e^{j\omega}) e^{-j\omega \frac{M}{2}} = A(e^{j\omega}) e^{-2j\omega}$$

$$= \left[\sum_{k=0}^{\frac{M}{2}} a_k \cos(k\omega) \right] e^{-2j\omega} = H(e^{j\omega}) \quad , \quad a_0 = h\left[\frac{M}{2}\right], \quad a_k = 2 h\left[\frac{M}{2} - k\right]$$

$$\cos(\varphi) = \frac{e^{j\varphi} + e^{-j\varphi}}{2}$$

TYPE 2

$$M=5, L=6$$

EVEN LENGTH + SYMMETRIC

$$h[n] = \{h[0], h[1], h[2], h[3], h[4], h[5]\}$$

$$h[n] = \{h[0], h[1], h[2], h[2], h[1], h[0]\}$$

$$H(e^{j\omega}) = \sum_{n=0}^M h[n] e^{-j\omega n} = h[0] + h[1] e^{-j\omega} + h[2] e^{-2j\omega} + h[2] e^{-3j\omega} + h[1] e^{-4j\omega} + h[0] e^{-5j\omega}$$

$$= h[0] (1 + e^{-5j\omega}) + h[1] (e^{-j\omega} + e^{-4j\omega}) + h[2] (e^{-2j\omega} + e^{-3j\omega})$$

$$= e^{-\frac{5}{2}j\omega} \left[h[0] (e^{-\frac{5}{2}j\omega} + e^{\frac{5}{2}j\omega}) + h[1] (e^{-\frac{3}{2}j\omega} + e^{\frac{3}{2}j\omega}) + h[2] (e^{-\frac{1}{2}j\omega} + e^{\frac{1}{2}j\omega}) \right]$$

$$= \left[h[0] \cdot 2 \cos\left(\frac{5}{2}\omega\right) + h[1] \cdot 2 \cos\left(\frac{3}{2}\omega\right) + h[2] \cdot 2 \cos\left(\frac{1}{2}\omega\right) \right] e^{-\frac{5}{2}j\omega}$$

$$= A(e^{j\omega}) e^{-j\omega \frac{M}{2}}$$

$$= \left\{ \sum_{k=0}^{\frac{M-1}{2}} d_k \cos\left[\left(k + \frac{1}{2}\right)\omega\right] \right\} e^{-\frac{5}{2}j\omega} = H(e^{j\omega}), \quad d_k = 2 h\left[\frac{M-1}{2} - k\right]$$

TYPE 3

$$M=4, L=5$$

ODD LENGTH + ANTI-SYMMETRIC

$$h[n] = \{h[0], h[1], h[2], h[3], h[4]\}$$

$$h[n] = \{h[0], h[1], h[2], -h[1], -h[0]\}$$

$$H(e^{j\omega}) = \sum_{n=0}^{M/2} h[n] e^{-j\omega n} = h[0] + h[1] e^{-j\omega} + h[2] e^{-2j\omega} - h[1] e^{-3j\omega} - h[0] e^{-4j\omega}$$

$$= h[0] (1 - e^{-4j\omega}) + h[1] (e^{-j\omega} - e^{-3j\omega}) + h[2] e^{-2j\omega}$$

$$= e^{-2j\omega} \left[h[0] (e^{2j\omega} - e^{-2j\omega}) + h[1] (e^{j\omega} - e^{-j\omega}) + h[2] \right]$$

$$= \left[h[0] \cdot 2j \sin(2\omega) + h[1] \cdot 2j \sin(\omega) + h[2] \right] e^{-2j\omega}$$

NOTE THAT SINCE THE FILTER IS ANTI-SYMMETRIC, WE HAVE THAT

$$h[n] = -h[M-n]$$

$$M=4$$

$$h[0] = -h[4]$$

$$h[1] = -h[3]$$

$$h[2] = -h[2] \rightarrow \text{THIS IS ONLY POSSIBLE IF } h[2] = 0$$

NOT IN TYPE 4 SINCE SYMMETRY AXIS IS BETWEEN TWO SAMPLES

IN FACT, IN TYPE 3 FILTERS THE CENTER VALUE MUST ALWAYS BE ZERO

$$= \left[h[0] \cdot 2j \sin(2\omega) + h[1] \cdot 2j \sin(\omega) \right] e^{-2j\omega}$$

$$= j \left[h[0] \cdot 2 \sin(2\omega) + h[1] \cdot 2 \sin(\omega) \right] e^{-2j\omega}$$

$$= jA(e^{j\omega}) e^{-j\omega \frac{M}{2}} = j \left[\sum_{k=0}^{M/2} d_k \sin(k\omega) \right] e^{-2j\omega} = H(e^{j\omega}) \quad , \quad d_0 = 0, \quad d_k = 2h\left[\frac{M}{2} - k\right]$$

$$= e^{j\frac{\pi}{2}} A(e^{j\omega}) e^{-j\omega \frac{M}{2}} = A(e^{j\omega}) e^{-j\omega \frac{M}{2} + j\frac{\pi}{2}}$$

TYPE 4

$M=5, L=6$

EVEN LENGTH + ANTI-SYMMETRIC

$$h[n] = \{h[0], h[1], h[2], h[3], h[4], h[5]\}$$

$$\downarrow$$

$$h[n] = \{h[0], h[1], h[2], -h[2], -h[1], -h[0]\}$$

$$H(e^{j\omega}) = \sum_{n=0}^M h[n] e^{-j\omega n} = h[0] + h[1] e^{-j\omega} + h[2] e^{-2j\omega} - h[2] e^{-3j\omega} - h[1] e^{-4j\omega} - h[0] e^{-5j\omega}$$

$$= h[0] (1 - e^{-5j\omega}) + h[1] (e^{-j\omega} - e^{-4j\omega}) + h[2] (e^{-2j\omega} - e^{-3j\omega})$$

$$= e^{-\frac{5}{2}j\omega} \left[h[0] (e^{\frac{5}{2}j\omega} - e^{-\frac{5}{2}j\omega}) + h[1] (e^{\frac{3}{2}j\omega} - e^{-\frac{3}{2}j\omega}) + h[2] (e^{\frac{1}{2}j\omega} - e^{-\frac{1}{2}j\omega}) \right]$$

$$= e^{-\frac{5}{2}j\omega} \left[h[0] \cdot 2j \sin\left(\frac{5}{2}\omega\right) + h[1] \cdot 2j \sin\left(\frac{3}{2}\omega\right) + h[2] \cdot 2j \sin\left(\frac{1}{2}\omega\right) \right]$$

$$= j e^{-\frac{5}{2}j\omega} \left[h[0] \cdot 2j \sin\left(\frac{5}{2}\omega\right) + h[1] \cdot 2j \sin\left(\frac{3}{2}\omega\right) + h[2] \cdot 2j \sin\left(\frac{1}{2}\omega\right) \right]$$

$$= e^{j\frac{\pi}{2}} e^{-\frac{5}{2}j\omega} \left[h[0] \cdot 2j \sin\left(\frac{5}{2}\omega\right) + h[1] \cdot 2j \sin\left(\frac{3}{2}\omega\right) + h[2] \cdot 2j \sin\left(\frac{1}{2}\omega\right) \right]$$

$$= \left[\sum_{k=0}^{\frac{M-1}{2}} d_k \sin\left(k + \frac{1}{2}\omega\right) \right] e^{-\frac{5}{2}j\omega + j\frac{\pi}{2}} \quad , \quad d_k = 2h\left[\frac{M-1}{2} - k\right]$$

$$= A(e^{j\omega}) e^{-j\omega \frac{M}{2} + j\frac{\pi}{2}}$$

EX - TYPE 1



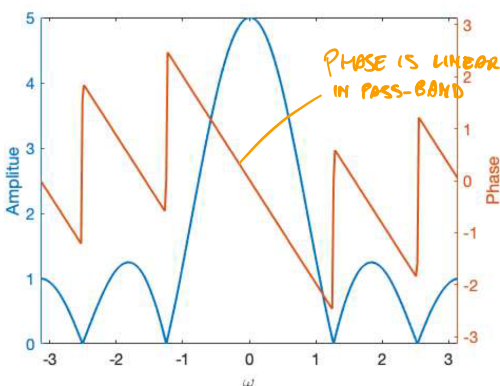
$$h[n] = u[n] - u[n-5]$$

$$H(z) = \frac{1}{1-z^{-1}} - \frac{z^{-5}}{1-z^{-1}} = \frac{1-z^{-5}}{1-z^{-1}}$$

$$\downarrow z=e^{j\omega}$$

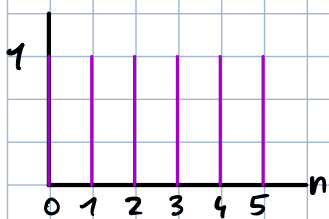
$$H(e^{j\omega}) = \frac{1-e^{-5j\omega}}{1-e^{-j\omega}} = \frac{e^{-\frac{5}{2}j\omega} (e^{\frac{5}{2}j\omega} - e^{-\frac{5}{2}j\omega})}{e^{-\frac{1}{2}j\omega} (e^{\frac{1}{2}j\omega} - e^{-\frac{1}{2}j\omega})} = e^{-\frac{5}{2}j\omega} e^{\frac{1}{2}j\omega} \frac{\sin(\frac{5}{2}\omega)}{\sin(\frac{1}{2}\omega)} = e^{-2j\omega} \frac{\sin(\frac{5}{2}\omega)}{\sin(\frac{1}{2}\omega)}$$

Low-Pass



THIS TERM CHANGES SIGN AT ZERO-CROSSINGS
LO PHASE SHIFTS BY $\pm\pi$

EX - TYPE 2



$$L=6$$

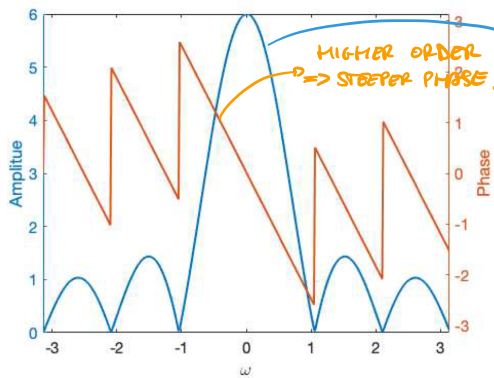
$$M=5$$

EVEN LENGTH
SYMMETRIC

$$h[n] = \mu[n] - \mu[n-6] \rightarrow H(z) = \frac{1}{1-z^{-1}} - \frac{z^{-6}}{1-z^{-1}} = \frac{1-z^{-6}}{1-z^{-1}}$$

$$H(e^{j\omega}) = \frac{1-e^{-6j\omega}}{1-e^{-j\omega}} = \frac{e^{-3j\omega}(e^{3j\omega}-e^{-3j\omega})}{e^{-\frac{1}{2}j\omega}(e^{\frac{1}{2}j\omega}-e^{-\frac{1}{2}j\omega})} = e^{-3j\omega} \frac{e^{\frac{1}{2}j\omega} \sin(3\omega)}{\sin(\frac{1}{2}\omega)} = e^{-\frac{5}{2}j\omega} \frac{\sin(3\omega)}{\sin(\frac{1}{2}\omega)}$$

Low-Pass



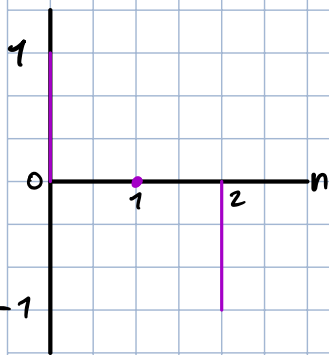
HIGHER ORDER
P -> STEEPER PHASE

LONGER FILTER -> NARROWER PASS-BAND

$$e^{j\omega \frac{M}{2}}$$

EX - TYPE 3

$$M=2, L=3$$

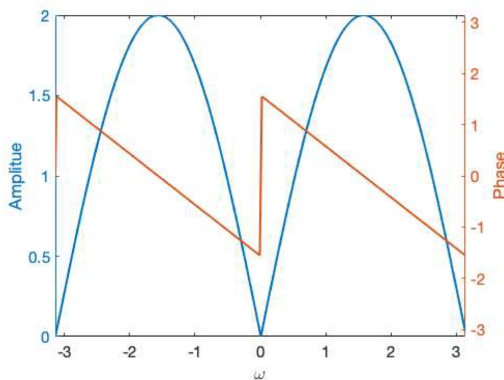


$$h[n] = \delta[n] - \delta[n-2] \rightarrow H(z) = 1 - z^{-2} \rightarrow H(e^{j\omega}) = 1 - e^{-2j\omega}$$

$$1 - e^{-2j\omega} = e^{-j\omega} (e^{j\omega} - e^{-j\omega}) = e^{-j\omega} \cdot 2j \sin(\omega) = 2 \sin(\omega) e^{j\omega + j\frac{\pi}{2}}$$

$$|H(e^{j\omega})| = |2 \sin(\omega)|$$

Band-Pass

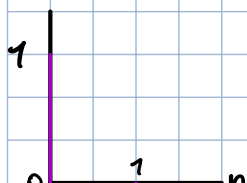


EX - TYPE 4

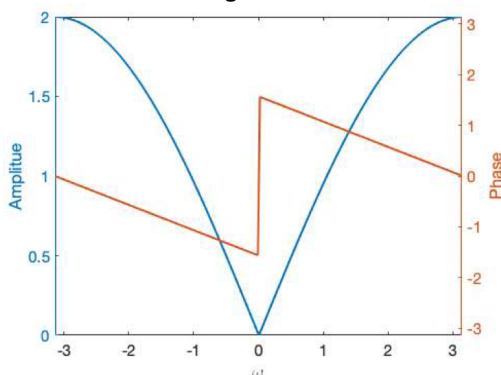
$$M=1, L=2$$

$$h[n] = \delta[n] - \delta[n-1] \rightarrow H(z) = 1 - z^{-1} \rightarrow H(e^{j\omega}) = 1 - e^{-j\omega}$$

$$1 - e^{-j\omega} = e^{-\frac{1}{2}j\omega} (e^{\frac{1}{2}j\omega} - e^{-\frac{1}{2}j\omega}) = e^{-\frac{1}{2}j\omega} \cdot 2j \sin(\frac{1}{2}\omega) = 2 \sin(\frac{1}{2}\omega) e^{-\frac{1}{2}j\omega + j\frac{\pi}{2}}$$



High-Pass



PROPERTIES OF ZEROS

TYPE 1 & 2

THEY HAVE IN COMMON THE FACT THAT THE IMPULSE RESPONSE IS SYMMETRIC

$$h[n] = h[M-n]$$

$$H(z) = \sum_{n=0}^M h[n] z^{-n} = \sum_{n=0}^M h[M-n] z^{-n} = \sum_{k=M}^0 h[k] z^{k-M} = z^{-M} \underbrace{\sum_{k=M}^0 h[k] z^k}_{H(z^{-1})} = z^{-M} H(z^{-1})$$

$$H(z) = z^{-M} H(z^{-1})$$

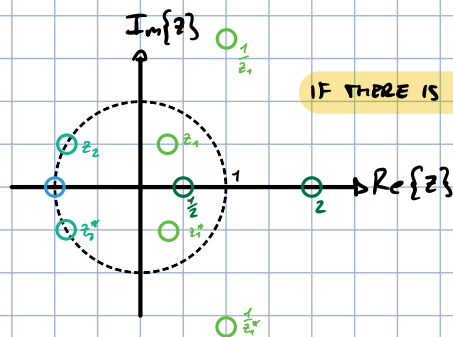
THIS PUTS A CONSTRAINT ON ZEROS

$$z = z_0 \rightarrow H(z_0) = 0 \Rightarrow H(z_0) = z_0^{-M} H\left(\frac{1}{z_0}\right) = 0 \Leftrightarrow \frac{1}{z_0} = 0$$

$\uparrow$
 z_0 is a zero of H

$\nwarrow$
 $\frac{1}{z_0}$ MUST BE A ZERO AS WELL

IF THE ZEROS ARE ON THE UNIT CIRCLE, THE ONLY CONSTRAINT IS SYMMETRY W.R.T. $\text{Re}\{z\}$



IF THERE IS A ZERO IN a , THERE MUST BE A ZERO IN $\frac{1}{a}$

IN THE PARTICULAR CASE WHEN M IS ODD (TYPE 2)

$$z = -1 \Rightarrow H(-1) = (-1)^M H\left(\frac{1}{-1}\right) \rightarrow H(-1) = (-1)^M H(-1)$$

AS LONG AS M IS EVEN THERE IS NO PROBLEM, BUT IF M IS ODD WE HAVE THAT $H(-1) = -H(-1)$, THEREFORE THE ONLY SOLUTION IS $H(-1) = 0$.

SO, IF M IS ODD, THERE IS A FIXED ZERO IN -1

TYPE 3 & 4

THEY HAVE IN COMMON THE FACT THAT THE IMPULSE RESPONSE IS ANTI-SYMMETRIC

$$h[n] = -h[M-n]$$

$$H(z) = \sum_{n=0}^M h[n] z^{-n} = - \sum_{n=0}^M h[M-n] z^{-n} = - \sum_{k=M-n}^0 h[k] z^{k-M} = - z^{-M} \sum_{k=M}^0 h[k] z^k = - z^{-M} H(z^{-1})$$

$$H(z) = - z^{-M} H(z^{-1})$$

DIFFERENCE IS SIGN

SUPPOSE WE HAVE A ZERO IN z_0 , THEN WE ALSO HAVE ZEROS IN $\frac{1}{z_0}$, z_0^* , $\frac{1}{z_0^*}$ AS BEFORE.

SUPP. $z=1$: $H(1) = (-1)^M H(1)$

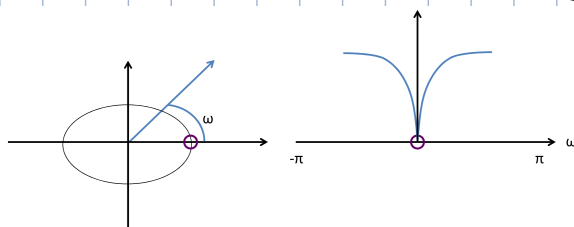
IN THIS CASE WE HAVE $H(1) = -H(1)$ REGARDLESS OF M . THE ONLY SOLUTION IS $H(1) = 0$, THEREFORE IN THE CASE OF TYPE 3 & 4 THERE IS ALWAYS A ZERO IN 1.

SUPP. $z=-1$: $H(-1) = -(-1)^M H(-1)$

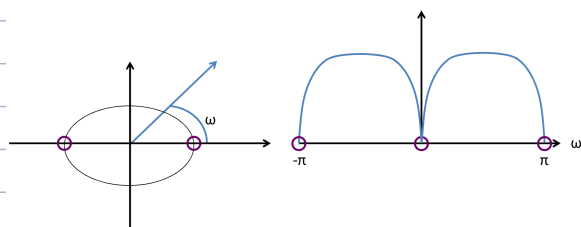
IN THIS CASE WE ALSO HAVE A FIXED ZERO IN $z=-1$ IF M IS EVEN / ODD (TYPE 3)

MEANING OF FIXED ZEROS

A PERMANENT ZERO IN $z=1$ IMPLIES IT'S IMPOSSIBLE TO DESIGN A LPF ($z=1/\omega=0$)



PERMANENT ZEROS IN $z=1$ AND $z=-1$ IMPLY IT'S IMPOSSIBLE TO DESIGN NEITHER A LPF NOR A HPF



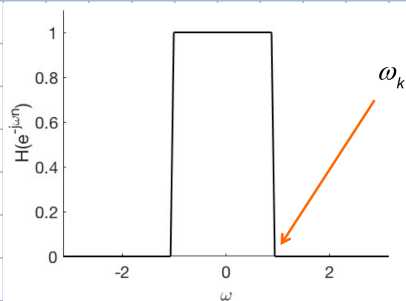
THERE ARE CONSTRAINTS ON THE DESIGN DEPENDING ON THE TYPE OF FILTER

FIR type/Design possibilities	Low Pass	High Pass	Band Pass	Stop Band
Type 1	Yes	Yes	Yes	Yes
Type 2	Yes	No (Permanent zero in $z=-1$)	Yes	No (Permanent zero in $z=-1$)
Type 3	No (Permanent zero in $z=1$)	No (Permanent zero in $z=-1$)	Yes	No (Permanent zero in $z=-1$ and $z=1$)
Type 4	No (Permanent zero in $z=1$)	Yes	Yes	No (Permanent zero in $z=1$)

WINDOWING

WE CAN DESIGN AN FIR FILTER STARTING FROM AN IDEAL ONE (IIR) AND WINDOWING (TRUNCATING IN TIME). THE PROCESS CONSISTS OF CHOOSING THE DESIRED IDEAL FILTER IN THE FREQUENCY DOMAIN, PERFORM THE ANTI-TRANSFORM TO GO IN THE TIME DOMAIN, WINDOW THE RESPONSE AND RE-TRANSFORM TO EVALUATE THE EFFECTS OF TRUNCATION.

LET'S CONSIDER A LPF



$$H(e^{j\omega}) = \begin{cases} 1 & |\omega| < \omega_k \\ 0 & \text{OTHERWISE} \end{cases} \quad \omega \in [-\pi, \pi]$$

IDEAL LPF

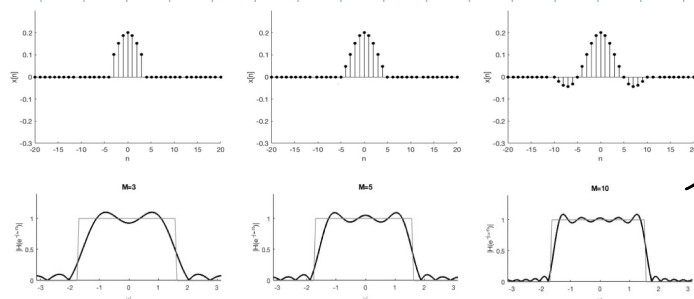
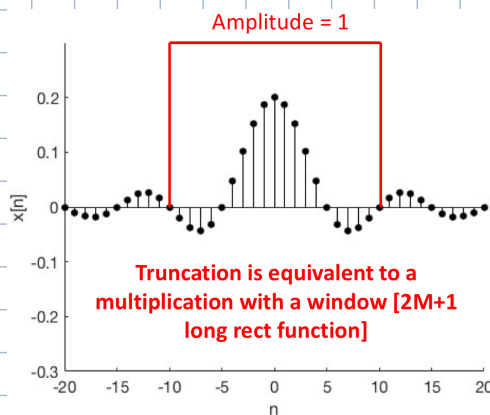
$$\begin{aligned} h[n] &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_k}^{\omega_k} e^{j\omega n} d\omega = \frac{1}{2\pi} \left. \frac{e^{j\omega n}}{jn} \right|_{-\omega_k}^{\omega_k} = \frac{1}{2\pi} \frac{1}{jn} (e^{j\omega_k n} - e^{-j\omega_k n}) = \\ &= \frac{1}{2\pi} \frac{1}{jn} 2j \sin(\omega_k n) = \frac{\sin(\omega_k n)}{\pi n} \end{aligned}$$

NOW WE GO BACK IN FREQUENCY DOMAIN AND WINDOW.

$$H(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} h[n] e^{-j\omega n} = \sum_{n=-\infty}^{+\infty} \frac{\sin(\omega_k n)}{\pi n} e^{-j\omega n} \xrightarrow{\text{WINDOW}} H_M(e^{j\omega}) = \sum_{n=-M}^M \frac{\sin(\omega_k n)}{\pi n} e^{-j\omega n}$$

WINDOWING IS A MULTIPLICATION OF THE IMPULSE RESPONSE WITH A WINDOWING FUNCTION (E.G. RECT) AND THEREFORE A CONVOLUTION IN FREQUENCY.

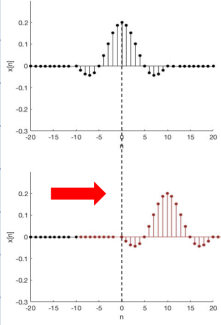
$$h_M[n] = h[n] w[n] \longleftrightarrow H_M(e^{j\omega}) = \frac{1}{2\pi} H(e^{j\omega}) * w(e^{j\omega})$$



LONGER FILTERS HAVE SHARPER TRANSITION BANDS, BECAUSE $\Pi[n] \rightarrow \text{sinc}(\omega)$ HAS NARROWER AND SHARPER MAIN LOBE

CAUSAL

THESE FILTERS INCLUDE PAST VALUES, TO MAKE THEM CAUSAL WE JUST NEED TO SHIFT



$$h_M[n] := h[n] w[n]$$

$$h_M^{\text{causal}}[n] := h_M[n-M]$$

RECTANGULAR WINDOW

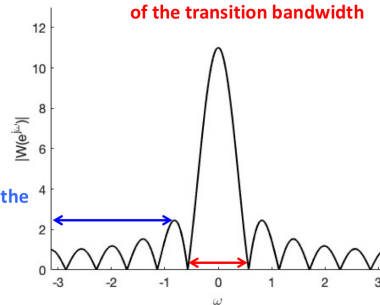
$$w[n] = \begin{cases} 1 & 0 \leq n \leq K \\ 0 & \text{ELSE} \end{cases}$$

$$W(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} w[n] e^{-j\omega n} = \sum_{n=0}^K 1 \cdot e^{-j\omega n} = \sum_{n=0}^K (e^{-j\omega})^n = \frac{1 - e^{-j\omega(K+1)}}{1 - e^{-j\omega}} = e^{-j\frac{\omega}{2}K} \frac{\sin\left[\frac{\omega(K+1)}{2}\right]}{\sin\left(\frac{\omega}{2}\right)}$$

Side lobes amplitudes determines the ripples amplitude

GIBBS PHENOMENON

Main lobe width determines the size of the transition bandwidth



GIBBS PHENOMENON

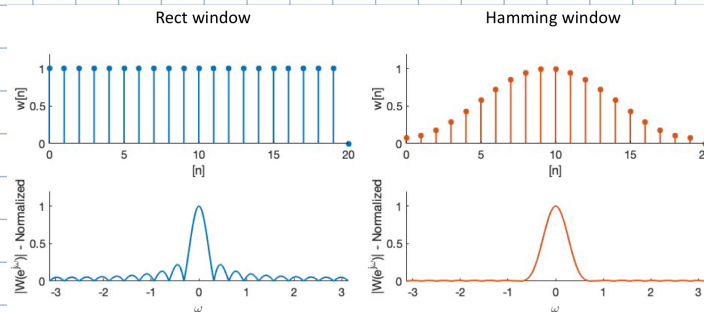
IT'S THE OSCILLATORY BEHAVIOR IN THE AMPLITUDE RESPONSE CAUSED BY THE TRUNCATION OF IDEAL FILTERS.

IF THE LENGTH OF THE FILTER INCREASES, THE NUMBER OF RIPPLES INCREASES AND THUS THE RIPPLE WIDTHS DECREASE.

THE HEIGHT OF THE RIPPLES (ENVELOPE) DOES NOT CHANGE.

GIBBS PHENOMENON CAN BE REDUCED USING SMOOTHER WINDOWS.

HAMMING WINDOW



THE MAIN LOBE IS WIDER, THEREFORE THE TRANSITION BAND IS LARGER, BUT THE SIDE LOBES ARE MUCH LOWER, RESULTING IN HIGHER ATTENUATION IN THE STOP-BAND AND REDUCED RIPPLE AMPLITUDE.